

Working notes: the zonality of SOC turnover

Storyline decision and diagnostic investigation of the prediction maps

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SUPERSEDED 2026-06-26 — historical record only. Going-forward planning has moved to `storyline_outline_2026-06-26.{tex,pdf}` (the story bible) and `results_reconciliation_2026-06-26.md` (number map + figure dispositions + manuscript checklist). This file is the 2026-06-25/26 exploration log; it is no longer maintained. Do not edit further.

Purpose. Capture the state of the discussion so we can resume cleanly. This documents (i) the chosen storyline, (ii) what we learned trying to build a global “zonality” map, (iii) a diagnostic investigation of the prediction pipeline (including two false alarms and one real issue), and (iv) where the story now stands and the next concrete step.

1 Storyline decision

We commit to **framing A**: the *zonality of SOC decay* as an *interpretive framework*, presented as the main contribution. The argument: SOC persistence is an emergent ecosystem property (Schmidt et al. 2011); if so, transit times (τ) carry a spatially structured residual *beyond climate*, analogous to classical pedogenic zonality. The deliverable that turns this from slogan into tool is an observation-constrained *map* of that beyond-climate modulation.

Supporting roles: the global τ map and CMIP6/ESM contrast are the empirical payload (B); the edaphic/mycorrhizal mechanism is the interpretive texture (C). The ESM comparison is *repositioned* from “validation” to “motivating illustration” (ESMs concentrate τ variation in a polar extremum and lack mid-latitude structure); this removes the dependence of the headline on the depth/pool comparability caveat, since the load-bearing claim becomes how spatial variation is *organised*, not its absolute magnitude.

2 Building the zonality map: what works and what fails

We must define “zone” *from the data*. We tested several constructions and scored each by how much of the **observed climate-residual** of $\ln \tau$ they explain (point level, out-of-bag; η^2 = between-zone variance fraction). Residual variance ≈ 0.32 (on the biological subset).

Why the intuitive map fails. The three non-climate channels (Edaphic, Land Use, Biological prediction shifts) are spatially collinear: a PCA of the signed $[E, L, B]$ adjustment vector gives $PC1 = 67\%$ with near-equal positive loadings (0.59, 0.53, 0.60; pairwise $r = 0.43\text{--}0.61$). The beyond-climate signal is therefore largely a *single shared modulation axis* (“how strongly non-climate processes bend τ away from the climate expectation”), not three independent process-geographies. This is a cleaner, more defensible framing than a process-mosaic — and it matches the data.

Recommended deliverable (3 layers):

Construction	explains observed climate-residual	verdict
Continuous RF modulation (M7–M1)	13.4% (ceiling)	validated signal
discretised, 5 zones	10.5%	works
discretised, 8 zones	11.5%	works
argmax “dominant process”	~0%	fails (collinear channels)
single hard-rule tree (CART)	~0%	fails (signal diffuse)
k -means on raw predictors	0%	fails

Table 1: Only a *supervised* construction works: discretising the validated continuous RF “non-climate modulation field” recovers most of the achievable signal at 5–8 zones. A Köppen-style “which process dominates here” map is *not* supported.

1. continuous global *modulation field* = primary zonality-intensity map;
2. a discrete 5–8 zone companion (quantile bins of the field) for communication;
3. per-zone driver characterisation (predictor profiles / ALE) restoring the “which process where” reading as an honest *secondary* layer.

3 Diagnostic investigation of the prediction maps

While preparing to build the map we stress-tested the global prediction pipeline (script 14). Net conclusion: **the maps are sound; the only real problem is coverage**. Two alarms were chased down and *refuted*.

Hypothesis	verdict
M1 climate raster is corrupted (mean 35.6 vs point 21.8 yr)	refuted — coverage artifact
SoilGrids / covariate scaling bug in script 14	refuted — all 19 predictors $r \approx 1.00$
Raster ($M7 - M1$) does not track the signal (-0.15)	refuted — in-sample extraction artifact
Microbial layers cover only 29% of land	confirmed — the real issue

Table 2: Outcome of the pipeline stress-test.

- **No M1 scaling bug.** The 35.6 vs 22.3 yr gap between M1 and M7 is a *domain* effect: M1 restricted to M7’s footprint = $23.8 \approx M7 = 22.3$. All six climate input rasters match the training ranges; offsets in central tendency are the temperate *sampling bias* of the training points.
- **No covariate scaling bug.** Per-predictor training-vs-raster check at 5,000 points: all 19 predictors (climate, SoilGrids edaphic with `d_factor`, terrain, microbial) have $r \approx 1.00$, median ratio ≈ 1.00 .
- **The -0.15 “inconsistency” was our own error:** comparing observations to *in-sample* raster predictions at the training points the models were fit on yields a meaningless in-bag residual. The correct out-of-bag comparison gives the non-climate adjustment correlating $+0.37$ with the observed climate-residual. *Any earlier raster-based $\eta^2 \approx 0$ numbers were contaminated by this same in-sample baseline; the point-level OOB results above stand.*

The real issue — coverage. M1 (climate) covers 2,152,154 land pixels; the microbial-dependent models (M4/M6/M7) cover only **631,252 = 29% of land** — the microbial layers (SPUN, Barceló, fungal proportion) are **NA** over $\sim 71\%$ of land (high latitudes, drylands). So the “global” τ map, the ESM comparison, and the planned zonality field inherit these holes precisely where the ESM-divergence story is strongest (boreal/Arctic). This is a data gap, not a code bug.

4 Option 2: M5 as the global-coverage workhorse

To get genuine global coverage we adopt **M5 (Climate + Edaphic + Land Use)** — no microbial dependency — as the main map, with M7 retained as a biologically-enriched overlay where microbial data exist. The M5 modulation field $\ln(M5/M1)$ is saved at `outputs/MRT_predictions/zonality_modul` (mean 0.018, SD 0.279, range $[-1.47, 2.01]$).

Model	land coverage	ΔR^2 vs climate (OOB)	explains climate-res
M5 (climate+edaphic+landuse)	49% (all soil-bearing land)	0.017	3.7%
M7 (+ biological)	29%	0.063	13.4%

Table 3: Coverage improves (29%→49%) but the abiotic signal is weak.

Key emergent finding. The *biological* (mycorrhizal) group carries most of the beyond-climate zonality, yet it is the least globally mapped. *The strongest beyond-climate signal is the least mappable.* Reframed positively, this is a result in its own right — soil biology is the dominant non-climate control on SOC turnover, and the absence of global soil-biology maps is the limiting factor. It motivates a concrete call for global microbial mapping and supports a **dual-layer deliverable**: a global *abiotic* zonality field (M5) plus a *biological enrichment* overlay (M7) where data permit.

5 Resolving process classes without a per-pixel “which-where” map

The channel collinearity blocks *one* thing, not everything. Keep two questions apart:

- **Q1** — “**which process class dominates *where*?**” (a per-pixel attribution map). **Blocked.** Channels correlate $r \approx 0.4$ – 0.6 , enough to make a pixel-by-pixel argmax unstable (the salt-and-pepper map). A Köppen-style process-territory map is *not* supportable.
- **Q2** — “**which process *class* carries beyond-climate signal, and how much?**” (aggregate). **Works.**

Key reconciliation: correlated $\neq$ redundant. Correlation ~ 0.5 is enough to break per-pixel argmax (Q1 fails) but *not* enough to make classes redundant: biology still adds large *unique* variance (M7 reaches 13.4% of the residual vs M5’s 3.7%). “Collinear” was too strong — the channels are correlated, not redundant.

Resolve Q2 by nested increments, not 3-way attribution. Use the nested ladder $M1 \rightarrow M5 \rightarrow M7$. Each increment is a *single* comparison, so its magnitude *and* geography are interpretable (unlike a simultaneous 3-way argmax):

- $M5 - M1 =$ **abiotic** modulation (edaphic + land use; land use is redundant, so $\approx$ edaphic). Globally mappable ($\sim 49\%$ /all soil).
- $M7 - M5 =$ **biological enrichment** (the *unique* biological contribution given abiotic + climate). Substantial and largely independent (since $M7 \gg M5$), on the 29% footprint.

Caveat: nested ΔR^2 is order-dependent (“biology given abiotic”); a commonality analysis (shared vs unique) is the rigorous version, but the headline (biology adds much beyond abiotic; land use redundant) is robust.

Coverage two levels = Tier 2. The coverage limit is not a bolt-on caveat; it *is* the scope boundary between the nested tiers: **Tier 1** (a beyond-climate zonality exists and is spatially organised) is the overarching thesis and does *not* depend on the 29% — it is visible in M5 over all soil-bearing land. **Tier 2** resolves that zonality into a global abiotic level ($M5 - M1$) and

a biological-enrichment zoom-in ($M7 - M5$, 29%). Narrative: a global Level-1 abiotic zonality, then a Level-2 deep-dive where soil biology is mapped. “Biological conclusions hold within the microbially-mapped domain” + a call for global soil-biology maps.

6 Routes to a clustered zone map

Two distinct jobs: (i) **discover** the zones (which pixels group together) and (ii) **describe** them with rules (portable, interpretable). Three constructions, increasing in richness. *Terminology: call the classes zones / turnover regimes, not “syndromes”.*

- **Route A — cluster the nested-increment space** [$M1$, $M5 - M1$, $M7 - M5$]. Cheap internal sanity check only; uses existing rasters. Collinearity-safe (increments, not raw channels).
- **Route C (alone) — regression tree on the modulation field.** One step: leaves are zones, splits are rules. Zones defined by *level* of modulation (intensity bands). Transparent, portable, Köppen-like.
- **Route B — cluster per-pixel driver contributions (TreeSHAP).** Each pixel gets a $\pm$ -years budget per predictor; clustering groups pixels shaped by the same driver combination. Richest for *insight*, but attribution has its own collinearity caveat and is not portable (needs the RF + SHAP to classify a new site). An analysis, not a tool.

The B→C bridge. B and C are not a forced chain — B *discovers*, C *describes*. To connect them, take B’s per-pixel cluster labels and fit a classification tree whose *response is the cluster ID* and whose inputs are the predictors. That tree emits a rule per zone (“if sand > 60% → zone 3; else if fungal > 0.2 & pH < 5 → zone 5”). Why bother over C-alone: a regression tree merges regions with the same τ -shift even if driven differently (a dryland shortened by sand vs by fungal + low-pH both read “−5 yr” → same leaf); B separates them by driver profile, C then names them. So B→C yields *mechanism-differentiated* zones; C-alone yields intensity bands.

When to use which. Build C-alone first (cheap). Only reach for B→C if intensity-only zones visibly lump mechanistically-different regions. If the modulation is $\sim$ 1-D (PCA PC1 = 67% suggests it may be), C-alone suffices and B adds little. **Recommendation:** lead with **C** as the flagship (portable rule-based zonation); use **B** to explain what each zone means; keep **A** as a throwaway check. Final manuscript object: a zonation map + a table giving each zone its (i) defining rule, (ii) characteristic τ -modulation, (iii) dominant drivers.

7 Caveats that must be resolved before drawing final figures

1. **OOB understates the signal.** The diagnostics above use OOB, where climate $R^2 = 0.53$; the manuscript uses *spatial CV*, where climate $R^2 = 0.28$. Spatial CV leaves a much larger residual for non-climate to explain, so the honest (spatial-CV) effect sizes are *larger* than the 3.7%/13.4% quoted here. **Re-validate the M5/M7 modulation fields with spatial-block CV** (reuse the existing 2° blocks).
2. **Edaphic ΔR^2 discrepancy.** Quick OOB gives edaphic+landuse only +0.017, whereas the manuscript reports edaphic-alone +0.076 (spatial CV). Almost certainly the OOB-vs-spatial-CV difference, but it is central to the variance-partition claims and must be reconciled explicitly.
3. **Map coherence.** At 0.1° the modulation field is spatially noisy, so quantile-bin zones fragment. Coherent (Köppen-like) zones require coarser resolution or spatially-constrained clustering.
4. **The 29% may be escapable.** It is the *intersection* of $\sim$ 9 biological layers. A “biology-light” model carrying only the single strongest biological variable — fungal proportion (Yu

2022, ~ 1 km, likely near-global) — would intersect far less and could push the biological tier toward $\sim 49\%$ while keeping the signal that mattered most in the ALE. **Check per-layer coverage first** (SPUN richness/endemism is the likely bottleneck).

5. **Biological footprint biased *within* the 29%.** Microbial data are dense in temperate Europe/N. America, thin in the tropics and high latitudes; biological conclusions are therefore doubly scoped.

8 Where we stand, and the next step

Standing: (i) storyline A is chosen and viable; (ii) the prediction pipeline is verified sound; (iii) the global zonality map is achievable as a continuous modulation field + discrete companion, but only *supervised* constructions work; (iv) the abiotic field (M5) is global-but-modest while the strong signal is biological-but-coverage-limited — now framed as a finding via a dual-layer (M5 global + M7 enrichment) deliverable.

Immediate next step (on reopen): set up **spatial-CV validation** of the M5 and M7 modulation fields, which simultaneously (a) yields paper-consistent effect sizes for the zonality signal and (b) reconciles the edaphic ΔR^2 against the manuscript’s $+0.076$. The magnitude this returns decides whether the zonality story is “modest but real” or “substantial,” i.e. how hard we can lean on framing A.

Then: (a) check per-layer biological coverage (is the 29% escapable via a fungal-proportion-only model?); (b) build the clustered zone map — lead with Route C (rule-based zonation) at a coherent resolution, add Route B (TreeSHAP contributions) to characterise each zone, with Route A as a sanity check; (c) decide whether to pursue microbial gap-filling (Option 1) as a sensitivity analysis to extend the biological tier globally. The deliverable is a global Level-1 abiotic zonality map + a Level-2 biological-enrichment zoom-in, each zone described by rule, τ -modulation, and dominant drivers.

9 Session update (2026-06-26): validation, scale, provenance, and a data-source correction

Spatial-block CV validation (common row set $n = 101,400$, 2° blocks, seed 42, 300 trees). All quantities computed on a single biological-complete row set so nested increments are strictly comparable (unlike the manuscript table, where each model used a different subset). Total $R_{CV}^2 = 0.378$.

- **Order-independent attribution (Shapley, sums exactly to total):** Climate **32.8%**, Edaphic **32.6%**, Biological **24.2%**, LandUse 10.4%. *Climate is only $\sim 1/3$ of the attributable signal; the non-climate domains carry $2/3$.* This is the headline and it is scale-independent.
- **Singles alone:** Edaphic $0.318 \geq$ Climate $0.306 >$ Biological $0.268 \gg$ LandUse 0.129 . Singles sum to 1.021 vs full $0.378 \Rightarrow$ overlap 0.64 (“correlated, not redundant”). Use *singles-first* as the Results opening.
- **Order-dependence reconciled:** biology added last (CELB–CEL) = $+0.008$ but Shapley share 24% — same data; LandUse redundant with Edaphic.
- **Modulation field is out-of-sample valid but modest:** $\text{cor}(\text{observed climate-residual, full modulation}) = 0.32$ ($R^2 = 0.10$); discrete 5–8 zones capture $\eta^2 \approx 0.07\text{--}0.09$. Biology is a spatially *separate* axis: $\text{cor}(\text{abiotic modulation, biological enrichment}) = -0.16$.

Scale spectrum and noise floor (appendix material). Residual variogram nugget/sill ≈ 0.66 : $\sim 2/3$ of the beyond-climate residual is unstructured noise (measurement + change-of-support + sub-reliable-resolution covariate error) at >15 km; the variogram is nearly pure-nugget beyond 15 km. η^2 /null shows coarse-scale organisation rising from $5\times$ (1°) to $33\times$ (20°) chance, but the absolute coarse signal is small ($\sim 3\text{--}7\%$ of variance), visible only after block-averaging. Covariates predict the *block-mean* (zonal) residual $2\text{--}3\times$ better than the point residual (R^2 $0.10 \rightarrow 0.20\text{--}0.29$, peaking at $\sim 3\text{--}4^\circ$). *Implication*: build the map at the coarse effective scale; never show native-pixel texture; the scale story is supporting (appendix), not the headline. Figure: `plots/appendix/scale_noisefloor.png`.

Provenance control — PASSES. Tested whether the coarse signal is a source/lab artifact using `source_db` (44 contributing databases). Only **2.4%** of beyond-climate residual variance is between sources; de-meaning by source leaves the coarse organisation almost intact (10° : $24\times \rightarrow 19\times$ chance) and covariate skill barely dented ($0.22 \rightarrow 0.19$ at 5°). The variogram mid-lag bump is unchanged by de-meaning, so it is *not* a between-source offset. Conclusion: the modest zonal signal is **ecological, not provenance**. (Caveat: source is confounded with geography, so this is a conservative/upper-bound test — surviving signal is definitely ecological.)

Data-source correction (important). The SOC data is **not** WoSIS. It is the OpenLandMap *Open Compendium of Soil Datasets: Soil Observations and Measurements* (Zenodo 15593990; `sol_chem.pnts.horizons.rds`). WoSIS is not even a `source_db` label in the data. Manuscript fixed in 4 places (§Soil data compilation, OC=0 sentence, Results data paragraph, Data availability) + new bib entry `OpenLandMapSoilDB2025` (author/year **TODO**: verify against Zenodo record). Source breakdown corrected: post-QC 198,382 profiles from **44** (not 47) databases; `GLanCE.points` (claimed 21.1% largest) is removed entirely by QC — the old paragraph applied the pre-QC breakdown to the post-QC count. Real top sources: USDA_NCSS 20.4%, Netherlands.BIS 11.5%, SISLAC 8.9%, CSIRO_NatSoil 7.0%, AfSPDB 6.7%.

Refined storyline. Lead with *attribution* (climate only $1/3$, robust and scale-free) and the ESM contrast: ESMs get zonality wrong in two independent ways — they *over-amplify* the climate range (polar $50\text{--}100\times$ vs data $\sim 3\text{--}5\times$; magnitude claim, caveated by steady-state/change-of-support) *and* omit the soil+biological axis that carries most of the real organisation (structural claim, robust). Bridge to next-generation models needing explicit mineral+microbial processes. Zonality is *real but modest*; present the map honestly at coarse scale.

10 Open issues / next steps

1. **Promote to production** (must-do before anything enters the MS): house-style `13c_commonality_validation` at 500 trees, writing `outputs/13c*.rds`, the singles $\rightarrow$ pairs $\rightarrow$ full + Shapley figure, the scale/variogram appendix figure, and the provenance-control table. Repeatable via `run_pipeline.R`.
2. **Biological-parsimony appendix (the “29% killer”)**. Check collinearity + per-layer coverage of the 9 biological layers; single-layer marginal ΔR^2 vs coverage; recommend one parsimonious layer to extend the biological footprint well beyond 29%. Figures $\rightarrow$ `plots/appendix/`; TODO comment already placed in `manuscript.tex` before **Supplementary tables**.
3. **Verify the compendium citation** (authors/year/version) from the Zenodo record; update `OpenLandMapSoilDB2025`.
4. **Add Moran’s I** on block-aggregated residuals as the standard spatial-autocorrelation statistic to accompany the η^2 /null backbone.

5. **Build the dual-layer zonality map** at the coarse effective scale ($\sim 3-4^\circ$), with per-zone driver characterisation (ALE/TreeSHAP).
6. **Reconcile MS magnitude numbers:** the current “ ~ 10 pp beyond climate” is partly a subset artifact; on comparable rows climate = 0.31 and beyond-climate ≈ 7 pp.
7. **Narrative:** write with Schimel OCAR; keep Jungian archetypes as a hidden internal scaffold only (deferred — do not implement in text).

11 Session update (2026-06-26, cont.): production decomposition, biological parsimony, Tier-1 Moran’s I

Validation promoted to house-style code. The exploratory spatial-CV scratchpad is now the production script `13c_commonality_validation.R` (500 trees, $2^\circ/10$ -fold/seed-42 block CV; env knobs `MRT_13C_BIO`, `MRT_13C_BLOCK`). It fits all 15 group coalitions on *one common-row set* so coalition R^2 are comparable and Shapley values sum exactly to the full-model R^2 . The all-9 sensitivity run reproduces the 300-tree memo numbers almost exactly (Climate 32.9, Edaphic 32.4, Biological 24.3, LandUse 10.4 %; full $R^2 = 0.378$; $n = 101,400$), validating the machinery.

Biological predictor decision (drop Barceló root colonization). Per-layer *observation* coverage is not uniform: `fungal_proportion` 99.8 %, the five SPUN richness/endemism layers 96.2 %, but the three Barceló root-colonization layers only ~ 62 % — the sole constraint collapsing the common-row set to $n = 101,400$. Equal-row block-CV (script `13d_biological_parsimony.R`) gives a unique Barceló-3 contribution of $+0.001 R^2$ (inert; alone -0.004). **MAIN model** therefore uses `fungal_proportion` + 5 SPUN layers (6 layers, 96 % coverage, ~ 84 % of the biological ΔR^2); the all-9 set is retained as a sensitivity case. Collinearity: only the derived ratios are redundant (`EcM_AM_root_ratio` $\approx$ `EcM_roots` 0.99; `EcM_AM_richness_ratio` $\approx$ `EcM_richness` 0.92); biology is otherwise multidimensional, so no single layer substitutes for the set.

Headline decomposition (MAIN, $n = 154,675$, full $R^2 = 0.380$). Dropping Barceló lifts the common-row set by 53 % with essentially no loss of full-model skill (a robustness point). Shapley shares: **Climate 35.1, Edaphic 32.8, Biological 22.3, LandUse 9.7 %** (sum exact) — climate $\approx \frac{1}{3}$, non-climate $\approx \frac{2}{3}$. Singles: $C\ 0.307 \approx E\ 0.307 > B\ 0.227 \gg L\ 0.110$; Σ singles = 0.950 vs full 0.380 $\Rightarrow$ overlap 0.57 (“correlated, not redundant”). Land-use unique ≈ 0 ; biology is badly *understated* by its nested-last increment ($+0.010$) versus Shapley (22 %) — **lead Results with Shapley**. Conclusion is insensitive to the biological-set choice (vs all-9: 32.9/32.4/24.3/10.4). Figure: `plots/step.13c_commonality/13c_decomposition.png`.

Tier-1 rigour — Moran’s I (script `13e_tier1_morans_i.R`). The beyond-climate residual (observed – climate-only block-CV prediction), aggregated to spatial blocks, has **Moran’s $I = 0.25$ at 2° and 0.26 at 5°** ($p \ll 0.001$): the residual is spatially *organised*, not noise, complementing the weak/scale-dependent ICC. (Cross-check: the climate-only block-CV $R^2 = 0.307$ here matches the 13c climate single exactly.) Moderate magnitude is consistent with the noise floor; zonality is real but modest.

Manuscript reconciliation (analysis done; prose pending discussion). The current “ ~ 10 percentage points beyond climate” (L826–827) is a *footprint artifact*: climate $R^2 = 0.280$ is scored on $n \approx 178k$ while the full model is on $n \approx 101k$. On a common footprint climate rises to 0.307 and the increment is ~ 7 pp. Full reconciliation table (old $\rightarrow$ new with line refs; mechan-

ical fixes vs. framing decisions) in `results_reconciliation_2026-06-26.md`. The qualitative ordering (edaphic largest non-climate, biology second, land-use redundant) survives.

Next. Run the $1^\circ/5^\circ$ block-size sweep (`MRT_13C_BLOCK`) for Shapley stability; then the Results-narrative discussion (Shapley-primary framing; replace/augment the M1–M7 table) and the manuscript edits.

Artifacts (2026-06-25/26): `outputs/MRT_predictions/zonality_modulation_M5_M1_logratio.tif`; `plots/appendix/scale_noise1`
exploratory scripts in the session scratchpad (`spatialcv_validation.R`, `spatialcv_finish.R`, `scale_noise1.R`,
`provenance_control.R`; earlier `zone_explore.R`, `zone_supervised.R`, `m5_zonality.R`). All reproducible from `outputs/12b_model_rea`
`outputs/13_rf_models.rds`, `outputs/13_var_groups.rds`. *Scratchpad scripts are exploratory and must be promoted*
to house-style pipeline code before any result enters the manuscript.